

A SHARPER CERTIFIED LOWER BOUND FOR ERDŐS' MINIMUM-OVERLAP PROBLEM

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ABSTRACT. We prove the computer-assisted lower bound

$$c_E > 0.380557$$

for Erdős' minimum-overlap constant. The proof builds on the 172-bin mean-conditioned dual-certificate framework used by L. Price to certify $c_E > 0.38055470$. Price's two central mean bins are replaced by a single even dual certificate using the same second-moment and pointwise Fourier bounds, together with the Parseval energy inequality already present in White's Fourier-analytic approach. The new center certificate has 69 nonzero rows. Its multipliers are treated as exact decimal rationals, and a standalone interval-arithmetic checker proves

$$\int_{-2}^2 q_+(t) dt \leq 2.6277192 < \frac{1}{0.380557}.$$

All noncentral bins in Price's published Arb verification have a larger margin; the worst of them has certified positive-part integral at most $2.627538530873375790\dots$. The optimizer used to discover the certificate is not part of the trusted proof path.

1. THE PROBLEM AND THE CONTINUOUS MODEL

For a partition $A \sqcup B = \{1, \dots, 2N\}$ with $|A| = |B| = N$, define

$$r_{A,B}(m) = |\{(a, b) \in A \times B : a - b = m\}|, \quad R(A, B) = \max_{m \in \mathbb{Z}} r_{A,B}(m),$$

and put

$$c_E = \liminf_{N \rightarrow \infty} \frac{1}{N} \min_{\substack{A \sqcup B = \{1, \dots, 2N\} \\ |A| = |B| = N}} R(A, B).$$

This is the asymptotic constant in Erdős' minimum-overlap problem. White's Fourier-analytic formulation and subsequent certified computations use the following continuous model [1, 2].

Let $\mathcal{H}$ be the set of measurable $h : [0, 2] \rightarrow [0, 1]$ with $\int_0^2 h = 1$. Put $g = 1 - h$ on $[0, 2]$ and extend both functions by zero to $\mathbb{R}$. Define

$$p_h(t) = \int_{\mathbb{R}} h(x)g(x+t) dx, \quad -2 \leq t \leq 2.$$

Then $p_h \geq 0$, $\text{supp } p_h \subset [-2, 2]$, and $\int p_h = 1$. Set

$$c_{\text{cont}} = \inf_{h \in \mathcal{H}} \|p_h\|_{\infty}.$$

Lemma 1 (Discrete-to-continuous reduction). *One has $c_E \geq c_{\text{cont}}$.*

Proof. Fix a partition $A \sqcup B = \{1, \dots, 2N\}$. For $I_k = [(k-1)/N, k/N]$ define

$$h_N = \sum_{a \in A} \mathbf{1}_{I_a}, \quad g_N = \sum_{b \in B} \mathbf{1}_{I_b}.$$

Then $h_N \in \mathcal{H}$. If $\tau(u) = (1 - |u|)_+$, a direct convolution calculation gives

$$p_{h_N}(t) = \frac{1}{N} \sum_{a \in A, b \in B} \tau(Nt - (b - a)).$$

For fixed real x , at most two of the numbers $\tau(x - d)$, $d \in \mathbb{Z}$, are nonzero, and their sum is at most 1. Grouping the preceding sum by the integer $d = b - a$ therefore gives

$$p_{h_N}(t) \leq \frac{1}{N} \max_d |\{(a, b) \in A \times B : b - a = d\}| = \frac{R(A, B)}{N}.$$

Taking the supremum in t , then the minimum over partitions and finally the liminf proves the claim. $\square$

Hence it is enough to prove a strict lower bound for every admissible overlap density $p = p_h$.

2. THREE ANALYTIC CONSTRAINTS USED IN THE CENTER CERTIFICATE

Fix $h \in \mathcal{H}$, write $p = p_h$, and define its first moment

$$\mu = \int_{-2}^2 tp(t) dt.$$

If X has density h , the constraints $0 \leq h \leq 1$ and $\int h = 1$ imply $1/2 \leq \mathbb{E}X \leq 3/2$ by the elementary rearrangement principle (the first moment is minimized by $h = \mathbf{1}_{[0,1]}$ and maximized by $h = \mathbf{1}_{[1,2]}$). Since $\mathbb{E}Y = 2 - \mathbb{E}X$ for the complementary density g , it follows that $\mu = \mathbb{E}(Y - X) = 2 - 2\mathbb{E}X \in [-1, 1]$.

2.1. The second moment.

Lemma 2. *For every admissible p ,*

$$\int_{-2}^2 t^2 p(t) dt = \frac{2}{3} + \frac{\mu^2}{2}.$$

In particular, if $|\mu| \leq 1/320$, then

$$\int t^2 p(t) dt \leq B_2 := \frac{2}{3} + \frac{1}{2 \cdot 320^2}.$$

Proof. Let X, Y be independent random variables with densities h, g respectively. Then p is the density of $Y - X$. Since $h + g = 1$ on $[0, 2]$,

$$\mathbb{E}X + \mathbb{E}Y = \int_0^2 x dx = 2, \quad \mathbb{E}X^2 + \mathbb{E}Y^2 = \int_0^2 x^2 dx = \frac{8}{3}.$$

Moreover $\mu = \mathbb{E}(Y - X)$, hence $\mathbb{E}X = 1 - \mu/2$ and $\mathbb{E}Y = 1 + \mu/2$. Independence now gives

$$\mathbb{E}(Y - X)^2 = \frac{8}{3} - 2\mathbb{E}X \mathbb{E}Y = \frac{8}{3} - 2 \left(1 - \frac{\mu^2}{4}\right) = \frac{2}{3} + \frac{\mu^2}{2}.$$

$\square$

2.2. Pointwise Fourier upper bounds.

For real ξ define centered transforms

$$H(\xi) = \int_0^2 h(x) e^{i\xi(x-1)} dx, \quad G(\xi) = \int_0^2 g(x) e^{i\xi(x-1)} dx.$$

Let

$$C_\xi = \int_{-2}^2 p(t) \cos(\xi t) dt, \quad \text{sinc } \xi = \begin{cases} \sin \xi / \xi, & \xi \neq 0, \\ 1, & \xi = 0. \end{cases}$$

Then

$$H(\xi) + G(\xi) = 2 \text{sinc } \xi, \quad \int p(t) e^{i\xi t} dt = \overline{H(\xi)} G(\xi).$$

Lemma 3. *For every $\xi \in \mathbb{R}$,*

$$C_\xi \leq \text{sinc}^2 \xi.$$

Proof. Write $H - G = a + ib$ with $a, b \in \mathbb{R}$ and $\sigma = \text{sinc } \xi$. Since $H + G = 2\sigma$,

$$\overline{H}G = \sigma^2 - \frac{a^2 + b^2}{4} - i\sigma b.$$

Taking real parts gives the result. $\square$

2.3. White's Parseval energy constraint. The following is the form of White's Parseval constraint that is useful in the mean-bin certificate. We include the short derivation to fix normalization.

Lemma 4 (Parseval row). *For every positive integer M ,*

$$-\sum_{n=1}^M C_{n\pi} \leq \frac{1}{2}.$$

Proof. At $\xi = n\pi$ with $n \neq 0$, one has $H(\xi) + G(\xi) = 0$, hence

$$C_{n\pi} = \operatorname{Re}(\overline{H(n\pi)}G(n\pi)) = -|H(n\pi)|^2.$$

The functions $e^{in\pi u}/\sqrt{2}$, $n \in \mathbb{Z}$, form an orthonormal basis of $L^2([-1, 1])$. Parseval applied to $u \mapsto h(u+1)$ yields

$$\sum_{n \in \mathbb{Z}} |H(n\pi)|^2 = 2 \int_0^2 h(x)^2 dx \leq 2 \int_0^2 h(x) dx = 2.$$

Because $H(0) = 1$ and $|H(-n\pi)| = |H(n\pi)|$,

$$1 + 2 \sum_{n=1}^{\infty} |H(n\pi)|^2 \leq 2.$$

Thus $\sum_{n \geq 1} |H(n\pi)|^2 \leq 1/2$, and every finite truncation satisfies

$$-\sum_{n=1}^M C_{n\pi} = \sum_{n=1}^M |H(n\pi)|^2 \leq \frac{1}{2}.$$

□

Remark 5. *Lemma 4 is not claimed as a new inequality. It is the Parseval energy restriction already used by White [1], rewritten in the Fourier coefficients C_ξ of the overlap density. The contribution of the present certificate is to combine this global energy row with Price's mean-conditioned dual-certificate framework [2].*

3. THE DUAL-CERTIFICATE PRINCIPLE

Lemma 6 (Positive-part dual bound). *Suppose an admissible density p satisfies finitely many linear inequalities*

$$\int_{-2}^2 p(t)\phi_j(t) dt \leq B_j, \quad j = 1, \dots, r.$$

For arbitrary multipliers $\lambda_j \geq 0$, define

$$q(t) = 1 + \sum_{j=1}^r \lambda_j (B_j - \phi_j(t)), \quad q_+(t) = \max(q(t), 0).$$

Then

$$\|p\|_\infty \geq \frac{1}{\int_{-2}^2 q_+(t) dt}.$$

Proof. Since $\int p = 1$ and every multiplier is nonnegative,

$$\int pq = 1 + \sum_j \lambda_j \left(B_j - \int p\phi_j \right) \geq 1.$$

As $p \geq 0$,

$$1 \leq \int pq \leq \int pq_+ \leq \|p\|_\infty \int q_+.$$

□

4. A REPLACEMENT CERTIFICATE FOR THE TWO CENTRAL MEAN BINS

Price's certificate partitions the possible mean μ into 172 bins. Its unique worst pair is

$$I_- = [-0.003125, 0] = [-1/320, 0], \quad I_+ = [0, 0.003125] = [0, 1/320].$$

The certificate accompanying the present note replaces precisely these two bins. It uses only the following valid rows:

$$\begin{aligned} \phi_2(t) &= t^2, & B_2 &= \frac{2}{3} + \frac{1}{2 \cdot 320^2}, \\ \phi_\xi(t) &= \cos(\xi t), & B_\xi &= \text{sinc}^2 \xi, \end{aligned}$$

and, with $M = 100$,

$$\phi_P(t) = - \sum_{n=1}^{100} \cos(n\pi t), \quad B_P = \frac{1}{2}.$$

The first row is valid on both bins by Lemma 2, the second by Lemma 3, and the third by Lemma 4.

The machine-readable file `certificate/center_certificate.json` contains 69 nonzero multipliers: one second-moment multiplier, 67 cosine multipliers, and one Parseval multiplier. Decimal strings are interpreted as exact rational numbers. All multipliers are nonnegative. The resulting function has the explicit form

$$\begin{aligned} q(t) &= 1 + \lambda_2(B_2 - t^2) + \sum_{\xi \in \Xi} \lambda_\xi(B_\xi - \cos(\xi t)) \\ &\quad + \lambda_P \left(\frac{1}{2} + \sum_{n=1}^{100} \cos(n\pi t) \right), \end{aligned}$$

where Ξ and all multipliers are exactly those in the JSON file. In particular, q is even, so a single certificate covers both I_- and I_+ .

4.1. Rigorous evaluation of the positive part. The checker `code/verify_center_chunked.py` does not trust the LP solver that found the multipliers. It reconstructs the certificate from the JSON file and verifies the following facts independently.

For every cosine row it recomputes an interval enclosure of $\text{sinc}^2 \xi$ using `mpmath.iv` at 45 decimal digits, then rounds the upper endpoint upward again to a rational decimal. Replacing B_ξ by a larger number can only weaken the dual certificate, so this preserves validity. The t^2 and Parseval right-hand sides are rational and are evaluated exactly.

The checker writes q and its elementary antiderivative Q explicitly. It works on $[0, 2]$ and uses evenness at the end. On an interval $J = [a, b]$ with center c and radius r , interval arithmetic encloses $q(c)$, $q'(c)$ and $q''(c)$, while a global interval bound M_3 encloses $\sup_{[-2,2]} |q'''|$. Taylor's theorem therefore gives the certified enclosure

$$q(c) - |q'(c)|r - \frac{|q''(c)|}{2}r^2 - \frac{M_3}{6}r^3 \leq q(x) \leq q(c) + |q'(c)|r + \frac{|q''(c)|}{2}r^2 + \frac{M_3}{6}r^3$$

for every $x \in J$, with every displayed numerical quantity replaced in the program by an outward interval bound.

If the certified upper endpoint is nonpositive, J contributes zero. If the certified lower endpoint is positive, $q_+ = q$ on J and the checker adds an outward-rounded upper bound for $Q(b) - Q(a)$. Otherwise it bisects J ; once the width is at most $2 \cdot 10^{-5}$, it conservatively charges the entire cell by $(b - a) \max(0, U_J)$, where U_J is the certified upper bound for q on the cell. A floating-point root search is used only to propose an initial subdivision and has no logical role: a missed root is still covered by the interval test and bisection.

A fresh 45-digit run on the files archived with this note terminated with

```
FINAL Dupper 2.6277191078658615742268756
TARGET 2.6277272524221075949200776756175815975...
MARGIN 0.0000081445562460206932020756175815975...
CERTIFIED True
```

The transcript above is a verbatim record of one run. Only the inequality is asserted, not the individual digits of **Dupper**: the initial subdivision is proposed by a floating-point root search, so its cell boundaries, and hence the trailing digits of the accumulated bound, depend on the version of SciPy in use. Runs on other library versions reproduce the cell counts and the certified inequality while differing in the last few digits. Accordingly we record the conclusion in the rounded form

$$(1) \quad \int_{-2}^2 q_+(t) dt \leq 2.6277192 < \frac{1}{0.380557}.$$

By Lemma 6, every admissible overlap density whose mean belongs to $I_- \cup I_+$ satisfies

$$\|p\|_\infty > 0.380557.$$

5. THE NONCENTRAL BINS AND THE GLOBAL THEOREM

For all other mean bins we reuse Price's published Arb-certified bounds [2]. The two old central rows (indices 85 and 86) are the only ones whose certified positive-part integrals exceed the threshold needed here. Among all remaining bins, the largest published certified value occurs at bin 77 (and, by symmetry, bin 94):

$$(2) \quad D_{\text{noncentral}} \leq 2.6275385308733757900902721758784 \dots$$

In particular,

$$\frac{1}{D_{\text{noncentral}}} \geq 0.3805843333028524 \dots > 0.380557.$$

The neighboring bins 84 and 87 already have $D \leq 2.627336460420282 \dots$, so the replacement of bins 85–86 leaves no gap in the mean cover.

Theorem 7. *The Erdős minimum-overlap constant satisfies*

$$\boxed{c_E > 0.380557}.$$

Proof. Let $h \in \mathcal{H}$ and $p = p_h$. Its mean μ lies in $[-1, 1]$, which is covered by Price's 172 mean bins. If $\mu \in I_- \cup I_+$, the new center certificate and (1) give $\|p\|_\infty > 0.380557$. Otherwise, Price's certified noncentral bin containing μ , together with (2) and Lemma 6, gives the same strict lower bound. Hence $c_{\text{cont}} > 0.380557$. Lemma 1 yields $c_E \geq c_{\text{cont}} > 0.380557$. $\square$

6. REPRODUCIBILITY AND TRUST MODEL

The new mathematical content that must be checked beyond the cited Price certificate is small: Lemmas 2, 3, 4, the dual Lemma 6, and the finite center certificate. The optimization procedure that discovered the multipliers is not needed to verify the theorem.

The trusted path for the center calculation consists of:

- exact Python integer and `Fraction` arithmetic for rational data;
- arbitrary-precision interval operations, including interval sine and cosine, in `mpmath.iv`;
- the finite subdivision and Taylor argument implemented in the checker.

Ordinary floating-point root finding and the LP solver are explicitly outside the trusted path. The repository records SHA-256 hashes for the certificate, checkers, paper source and verification transcripts in its root `SHA256SUMS.txt`, and a second manifest under `vendor/price/` covers the redistributed upstream package. Both are checked in continuous integration, and the repository disables end-of-line translation so that a clone reproduces the hashed bytes on every platform.

As a second software implementation, the release also contains `code/verify_center_mpfr.c`. This checker calls MPFR directly with explicit directed rounding, uses no `mpmath`, NumPy, SciPy, Arb, or root finder, and starts from a dyadic subdivision of $[0, 2]$. It certifies signs using an independent quadratic Taylor enclosure and evaluates positive cells with its own outward-rounded antiderivative calculation. At both 256-bit and 384-bit precision it proves

$$\int_{-2}^2 q_+(t) dt \leq 2.62772268405113257247485152716856 \dots < \frac{1}{0.380557},$$

and terminates with **CERTIFIED True**. Thus the new center certificate has been checked by two independent software implementations. This is not yet an external third-party audit; independent reproduction by another researcher remains desirable before using that stronger description. Use of AI.. Computational exploration, certificate search, and parts of the initial exposition were carried out with substantial assistance from OpenAI’s ChatGPT. The language model is not part of the trusted proof path. The human author is responsible for all claims made here.

REFERENCES

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